MIL-I-2781D
INTERIM AMENDMENT-4(SHIPS)
9 January 1973

SUPERSEDING 1/
INTERIM AMENDMENT-3(SHIPS)
1 March 1971

MILITARY SPECIFICATION

INSULATION, PIPE, THERMAL

This Interim Amendment is issued for use by the Naval Ship Engineering Center with Military Specification MIL-I-2781D dated 13 June 1963.

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1.2: Under Grade II, delete "Class c - Fibrous"; and Grade III, delete "Class f - Fibrous."

2.2: Delete all reference to "OFFICIAL CLASSIFICATION COMMITTEE" and substitute the following:

"UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, # Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)"

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3.1, Delete and substitute:

"3.1 Qualification. The thermal pipe insulation furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.2)."

3.2, delete and substitute:

"3.2 Material. The thermal pipe insulation shall be composed of asbestos-free and silica-free heat-resisting compounds suitable for the temperature conditions and the purpose intended."

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4.1, Delete and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

4.2, line 2: Delete "Bureau of Ships" and substitute "Naval Ship Engineering Center".

Footnote 1: After "Provisions Governing Qualification" add "SD-6".

1/ THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

FSC 5640

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Table III, delete and substitute:

"Table III - Physical requirements.

Characteristic	Test paragraph	Grade I Class b	Grade II Class d	Grade III Class e	
				Type I	Type II
Density average, lbs/cu.ft. maximum	4.4.2	14.0	14.0	14.0	22.0
Thermal conductivity (average) B.t.u. in. hr sq ft °Fahr. at mean temperature of:	4.4.3				
200°F.		.40	.40		
300°F.		.45	.45		
400°F.		.50	.50		
500°F.		---	---	.60	.66
600°F.		---	---	.65	.71
700°F.		---	---	.70	.76
Weight loss by tumbling Loss in weight (average), percent max.	4.4.4				
After first 10 minutes		40	40	40	55
After second 10 minutes		70	70	70	80
Modulus of rupture (average), lbs./sq.in., minimum	4.4.5	1/	1/	1/	2/
Changes under soaking heat for 6 hours at °F.	---	600	750	1,200	1,200
Linear shrinkage (average) percent, max.	4.4.6	2.0	2.0	2.0	2.0

1/ Three times density (lbs. per cu. ft.) of the sample tested.

2/ Two and one-half times density (lbs. per cu. ft.) of the sample tested."

4.3.1.1.1, line 3: Delete "I" and substitute "II", and delete "4.0 percent" and substitute "2.5 percent".

4.3.1.1.2, line 3: Delete "I-6" and substitute "II".

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Table IV, delete and substitute:

"Table IV - Sampling for tests.

Lot size in half section or segments	Sample size = Number of test specimens for each test (4.4.2, 4.4.4, 4.4.5, 4.4.6)	Number of test failures allowed (each test)
Up to 63	None	-
64 to 160	2	0
161 to 400	3	0
401 to 1,000	5	0
1,001 to 2,500	8	0
2,501 to 6,300	13	1
6,301 to 16,000	20	2
16,001 to 40,000	32	3

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